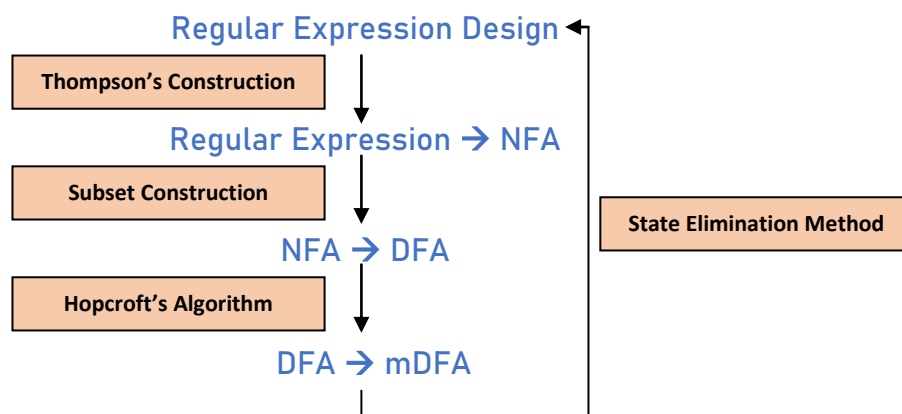


Lecture 11.1: Conversion of DFA to Regular Expression

Presenter: Sabrina Zaman Ishita (SZI)

Scribe: Sabrina Zaman Ishita (SZI)

In our previous lectures, we have shown you how to design different Regular Expressions, then we showed you how to convert regular expressions to NFA/ε-NFA through Thompson's Construction. Then you have been taught how to convert an NFA to DFA by subset-construction. We have further minimized this DFA by applying Hopcroft's Algorithm. Now, we will introduce you to the conversion mechanism of DFA to Regular Expression and we are going to use State Elimination Method to achieve this.



Following are some simple DFAs for which we can identify the Regular Expressions by simply looking at them:



But it is very difficult to figure out the Regular Expression of a DFA as given in Figure 1. For this we need follow algorithm. We are going to apply **State Elimination Method** to convert a DFA to its equivalent Regular Expression form.

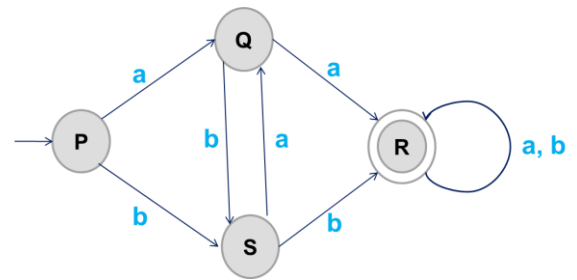
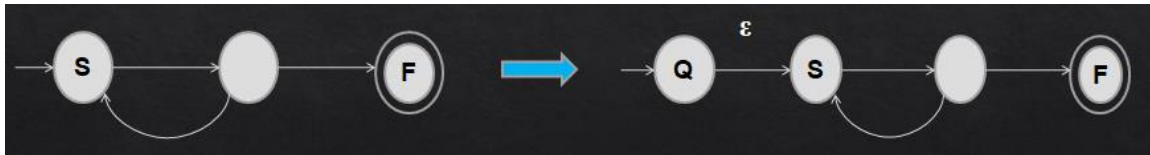


Figure 1:DFA

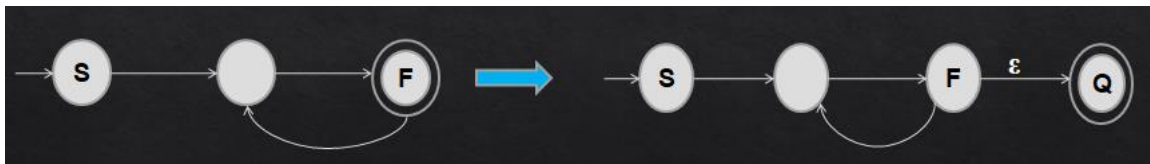
State Elimination Method:

Following rules are followed to apply this method:

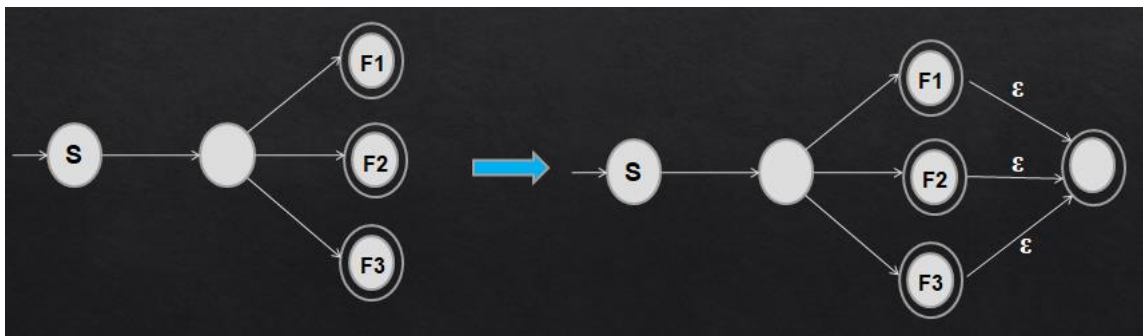
- ❑ If the start state has any incoming edge, create new start state having no incoming edge to it.



- ❑ If the final state has any outgoing edge, create new final state having no outgoing edge from it.

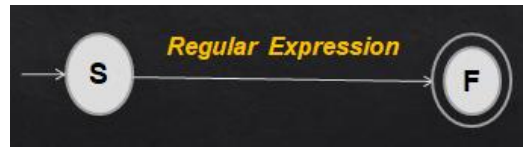


- ❑ If there are multiple final states, convert all the final states into one single final state.



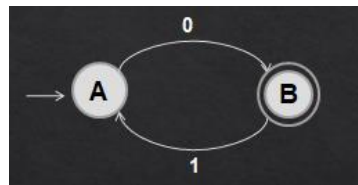
- ❑ Eliminate the intermediate states one by one (in any order) until the start state and final state exist.

❑ **Regular Expression** = cost of transition between start state and final state.

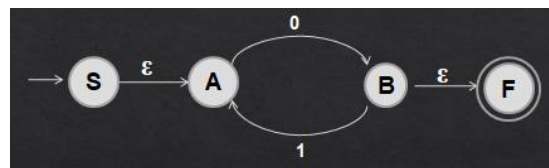


Apply State Elimination Method on an Example:

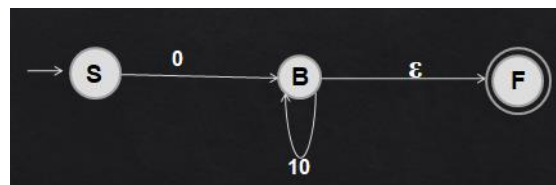
Example 1:



Let's add a new start state and a new final state as per the rules above.



Now, we need to eliminate state A and state B one by one in any order. Let's eliminate state A first and update the transitions which were through state A.



Now, let's eliminate state B which is very straightforward.



So, the equivalent Regular Expression for the given DFA is:

$0(10)^*$

If we had eliminated state B first, and the state A, we would have got different Regular Expression which is $(01)^*0$. But if you consider these regular expressions and try to figure out the strings they will be generating, you will get that they will be generating same set of strings.